

SHREYA CHENNUPATI

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Software Engineer specializing in **backend systems**, **distributed computing**, and **large-scale infrastructure**. Experienced in designing and deploying **production-grade services** using **Python, Go, Java, and C++** in **Linux** and **cloud environments**. Built **scalable REST APIs**, **distributed data pipelines**, and **ML inference platforms** supporting **500+ concurrent requests** and **high-volume datasets**. Strong foundation in **algorithms**, **distributed systems**, and **performance optimization** for reliable large-scale software systems.

EDUCATION

University of Florida, Gainesville, FL	08/2024 – Present
<i>Master of Science in Artificial Intelligence Systems</i>	GPA: 3.95/4.00

Keshav Memorial Institute of Technology, Telangana	08/2020 – 06/2024
<i>Bachelor of Technology in Computer Science and Engineering</i>	GPA: 3.92/4.00

SKILLS

Programming Languages: Python, Java, C++, C, Go

Backend Development: REST APIs, Microservices Architecture, Backend Systems Design, Distributed Systems

Databases: PostgreSQL, MySQL, MongoDB, Redis

Machine Learning: Deep Learning, Computer Vision, NLP, Model Training Pipelines

Cloud & DevOps: AWS (EC2, S3, Lambda), Docker, Kubernetes, CI/CD (Azure DevOps)

Developer Tools: Git, GitHub, Linux, Bash, Maven, GDB, Valgrind

Certifications: AWS Certified Cloud Practitioner, AWS AI Practitioner, Python PCAP Certification

Key Coursework: Distributed Systems, Large-Scale Storage Systems, High Performance Computing, Deep Learning, Computer Vision, Natural Language Processing

PROFESSIONAL EXPERIENCE

University of Florida	09/2024 – Present
<i>Graduate Software Engineer</i>	Gainesville, FL

- Designed and developed a **full-stack research platform** supporting visualization and analysis of large-scale genomic datasets using **Django, React, and PostgreSQL**.
- Designed scalable backend services and optimized PostgreSQL queries processing 100K+ records, reducing latency and improving system throughput.
- Implemented **secure authentication and RBAC authorization** enabling controlled multi-team access to research data.
- Developed automated **data ingestion and preprocessing pipelines** in Python and Bash for high-volume research datasets.
- Collaborated with interdisciplinary research teams to deploy scalable data processing systems supporting large scientific workloads.

Wislogix Consulting	05/2025 – 08/2025
<i>Software Developer Intern</i>	Irmo, SC

- Developed backend components for AI-driven outreach systems improving targeting accuracy by **12%**.
- Built scalable **REST APIs** supporting analytics pipelines and real-time engagement tracking.
- Designed automated data validation pipelines processing **1,000+ records**, eliminating inconsistencies and improving system reliability.
- Improved service deployment speed by **35%** through automation of backend personalization workflows.

Keshav Memorial Institute of Technology	05/2022 – 05/2024
<i>Student Researcher</i>	India, Telangana

- Conducted research on **machine learning-based detection of harmful online content** using large-scale comment datasets.
- Implemented data preprocessing pipelines in Python and trained classification models achieving **95.2% accuracy**.
- Investigated scalability limitations in ML moderation systems for high-volume social media platforms.

<i>Research Assistant</i>	India, Telangana
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- Developed high-performance C++ simulation modules to analyze institutional workflows processing 10K+ records.
- Optimized indexing logic using STL hash maps and dynamic arrays, improving lookup performance by 25%.
- Performed **memory profiling and debugging using GDB and Valgrind**, eliminating runtime leaks and reducing memory overhead.
- Conducted algorithm benchmarking and scalability analysis, decreasing system processing latency by **20%**.

PERSONAL PROJECTS

Scalable Machine Learning Inference Platform

01/2023 – 05/2023

Python, FastAPI, Docker, Redis, PostgreSQL

- Built containerized ML inference infrastructure using **FastAPI, Docker, Redis, and PostgreSQL**.
- Designed horizontally scalable backend services capable of handling **500+ concurrent API** requests using FastAPI and asynchronous processing.
- Implemented Redis caching to reduce API latency by **35%**.
- Developed automated testing pipelines ensuring reliable model deployment.

Taskify – Full-Stack Collaborative Task Platform

06/2023 – 10/2023

Go, React, PostgreSQL

- Designed full-stack project collaboration system using **Go, React, and PostgreSQL**.
- Developed backend microservices exposing REST APIs for task lifecycle management.
- Built real-time project dashboards enabling collaborative workflow management.

Driver Drowsiness Detection System

01/2024 – 03/2024

Computer Vision — Deep Learning

- Built computer vision pipeline using **deep learning models** trained on 10,000+ facial images.
- Developed real-time detection system improving inference speed by **25%**.

Hand Gesture Recognition System

10/2023 – 12/2023

Machine Learning — TensorFlow

- Built a CNN-based gesture classification pipeline achieving 86% accuracy through data augmentation and hyperparameter tuning.
- Refactored training workflows to enable reproducible experimentation and structured evaluation metrics.

LEADERSHIP & ACHIEVEMENTS

- 2023-24 hackathon Winner at KMIT, NGIT
- Head of Street Cause
- Two-time **All-India National Winner** (Dance)
- National-Level Choreographer